

Review on Linear Equations, Linear Graphs, Systems of Equations, and Equations with Parentheses (Homework)

① Identify the form of each linear equation below by writing "S" for standard, "SI" for slope-intercept, or "PS" for point-slope.

I) $3x - 2 = y$

Answers

SI

II) $-7x - 8y = -10$

S

III) $3 = x - y$

S

IV) $\frac{4}{3}(x+4) = y-11$

PS

V) $y+1 = 2(x+1)$

PS

VI) $3.5x = y$

SI

VII) $\frac{9x}{11} + \frac{y}{3} = -1$

S

VIII) $-(x-6) = y+5$

PS

IX) $y = \frac{x}{4}$

SI

X) $y = -\frac{3}{3}x$

SI

② Identify the slope and y-intercept of each of the following equations.

I) $y = 2x + 5$

$m=2$ $b=5$

II) $-3x = y$

$m=-3$ $b=0$

III) $y = \frac{1}{5}x - 1.3$

$m=\frac{1}{5}$ $b=-1.3$

IV) $-\frac{x}{3} - \frac{4}{7} = y$

$m=-\frac{1}{3}$ $b=-\frac{4}{7}$

V) $y = -\frac{4}{5}x + 101$

$m=-\frac{4}{5}$ $b=101$

③ Write linear equations with the following slopes and y-intercepts.

I) $m=-6$, $b=9$

$y=-6x+9$

II) $m=-\frac{3}{2}$, $b=0$

$y=-\frac{3}{2}x$

III) $m=0.45$, $b=\frac{1}{2}$

$y=0.45x+\frac{1}{2}$

IV) $m=-1$, $b=0.1$

$y=-x+0.1$

V) $m=7,000$, $b=7.7$

$y=7000x+7.7$

④ Find the slope of each of the following linear equations and a point on the line produced by graphing the equation.

I) $4(x+5) = y+11$

$m=4$, $(-5,-11)$

II) $y-1 = -2(x+7)$

$m=-2$, $(-7,1)$

III) $\frac{2}{9}(x-4) = y-5$

$m=\frac{2}{9}$, $(4,5)$

IV) $-(x+14) = y+5.99$

$m=-1$, $(-14,-5.99)$

V) $-5(x-6) = y$

$m=-5$, $(6,0)$

⑤ In each problem below, write an equation of the line with the given slope and point.

I) $m = -0.4, (3, 9)$

$$y - 9 = -0.4(x - 3)$$

II) $m = \frac{1}{2}, (-1, 2)$

$$y - 2 = \frac{1}{2}(x + 1)$$

III) $m = 26, (-3, -10)$

$$y + 10 = 26(x + 3)$$

IV) $m = -1, (7.4, -4.4)$

$$y + 4.4 = -(x - 7.4)$$

V) $m = 6, (3, 0)$

$$y = 6(x - 3)$$

A) Convert the following linear equations from standard form to slope-intercept form.

B) Then find the slope and y-intercept of the lines produced by graphing the equations.

⑥ $7x + y = 22$

A) $y = -7x + 22$ B) $m = -7, b = 22$

⑦ $-15x - 3y = 2$

A) $y = -5x - \frac{2}{3}$ B) $m = -5, b = -\frac{2}{3}$

⑧ $2x + 6y = -1$

A) $y = -\frac{1}{3}x - \frac{1}{6}$ B) $m = -\frac{1}{3}, b = -\frac{1}{6}$

⑨ $x + 2y = 3$

A) $y = -\frac{1}{2}x + \frac{3}{2}$ B) $m = -\frac{1}{2}, b = \frac{3}{2}$

A) In each problem below, use the slope equation to find the slope of the line that contains the two given points.

B) Then write the point-slope form of the equation of the line containing the two given points.

C) Then convert the equations from point-slope form to slope-intercept form.

⑩ $(4, 7) \& (2, -3)$

A) $m = 5$ B) $y - 7 = 5(x - 4)$

or $y + 3 = 5(x - 2)$

C) $y = 5x - 13$

⑪ $(-1, -2) \& (-6, 2)$

A) $m = -\frac{4}{5}$

B) $y + 2 = -\frac{4}{5}(x + 1)$

or $y - 2 = -\frac{4}{5}(x + 6)$

C) $y = -\frac{4}{5}x - \frac{14}{5}$

⑫ $(0, 0) \& (3, 1)$

A) $m = \frac{1}{3}$

B) $y - 0 = \frac{1}{3}(x - 0)$

or $y - 1 = \frac{1}{3}(x - 3)$

C) $y = \frac{1}{3}x$

⑬ $(-6, -1)$ & $(-2, -5)$

A) $m = -1$

B) $y + 1 = -(x + 6)$

or $y + 5 = -(x + 2)$

C) $y = -x - 7$

For problem #'s 14-34, the answers are on a different page.

Graph the following equations using method I, finding two random points on the line.

⑭ $y = 4x$

⑮ $5x - 3y = 18$

⑯ $-3 = 2x + 5y$

Graph the following equations using method II, finding the x and y intercepts.

⑰ $y = \frac{1}{2}x - 5$

⑱ $-5x + 2y = 2$

⑲ $x = \frac{y}{-2} + 1$

Graph the following equations using method III, slope-intercept form.

⑳ $y = 2x + 2$

㉑ $y = -x - 3$

㉒ $\frac{2}{3}x - 2 = y$

Graph the following equations using method IV, point-slope form.

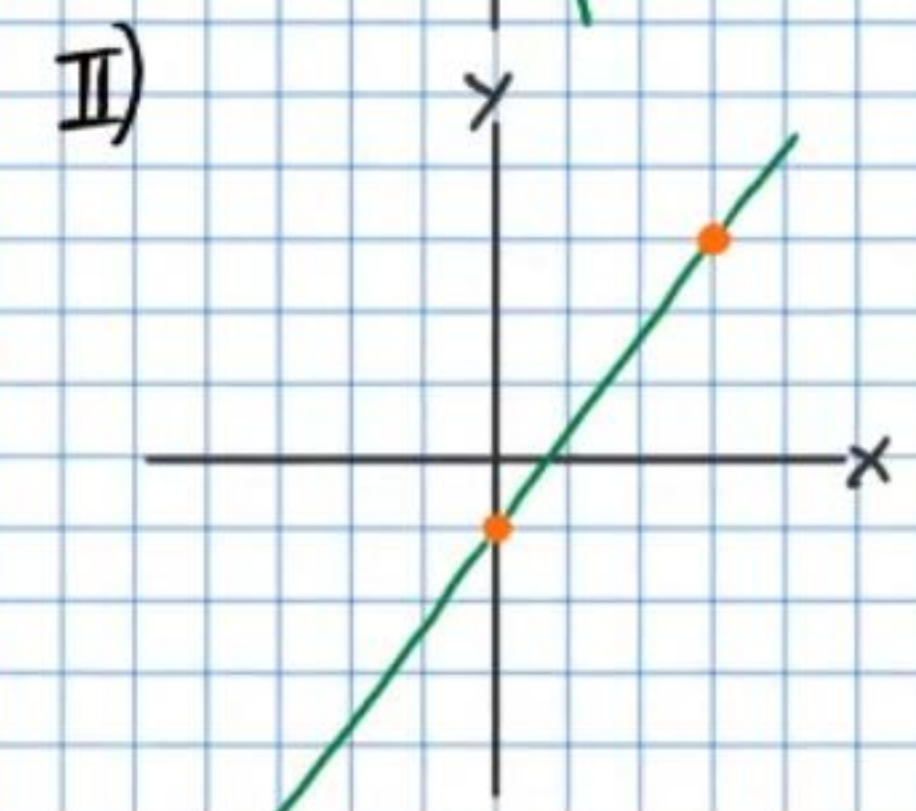
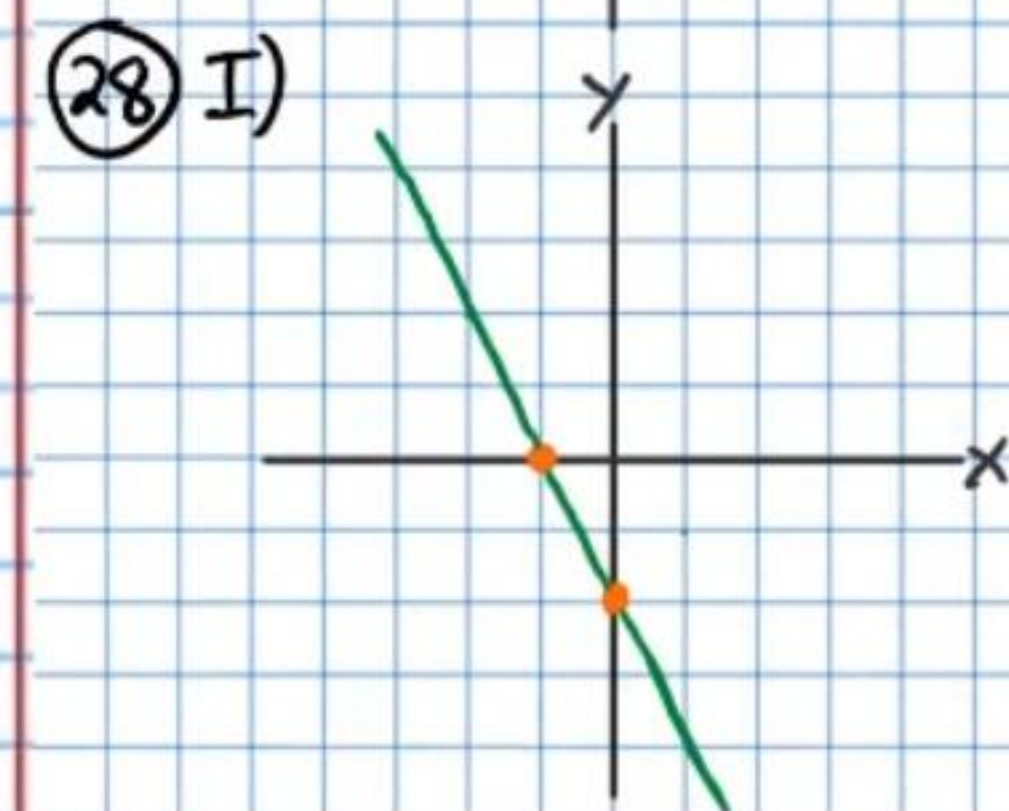
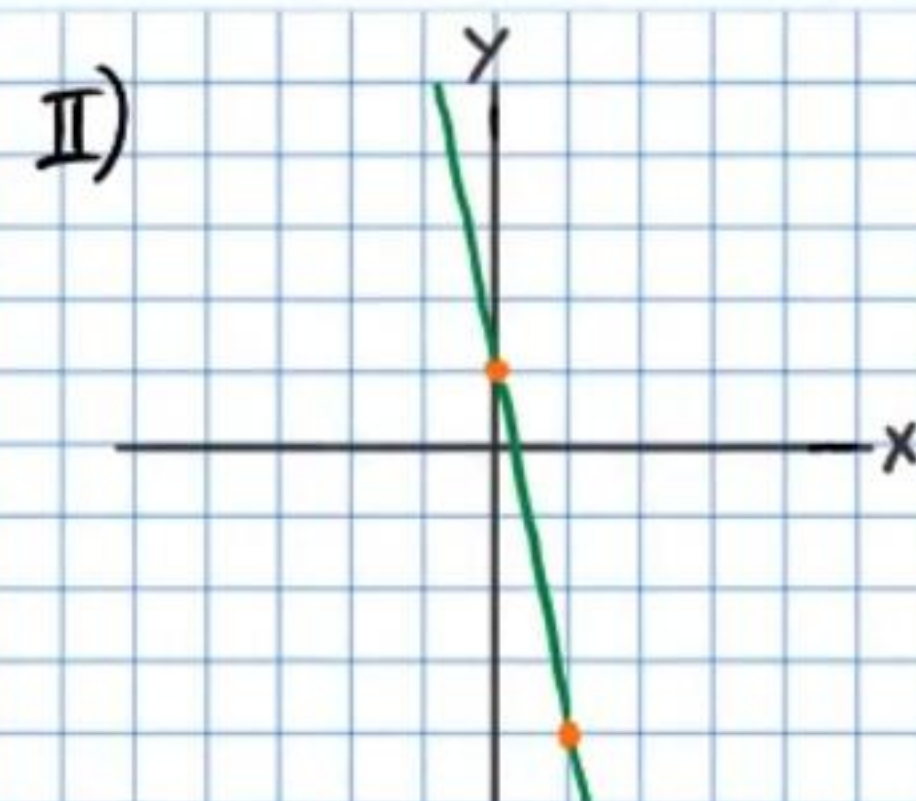
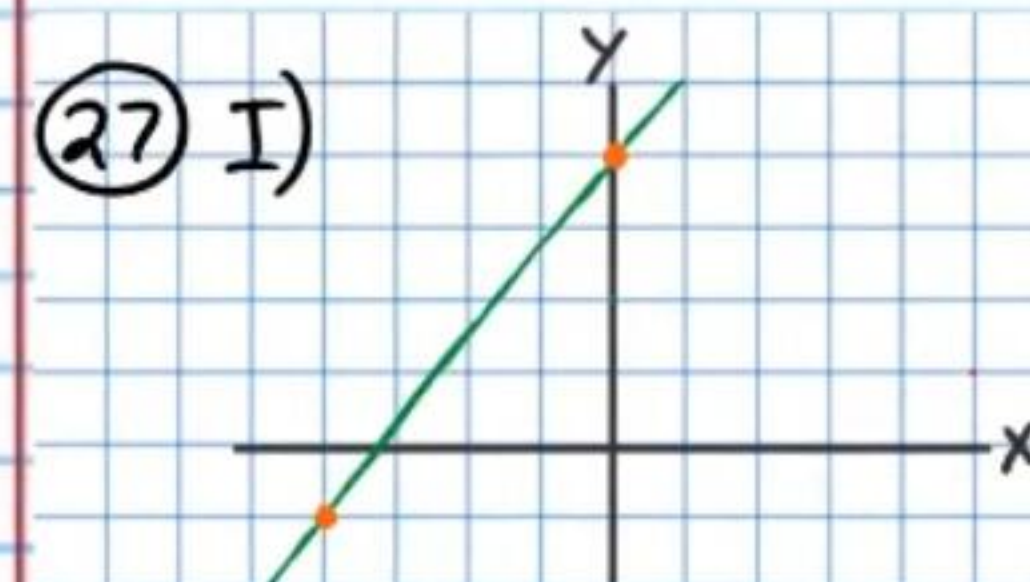
㉓ $y + 2 = -3(x - 1)$

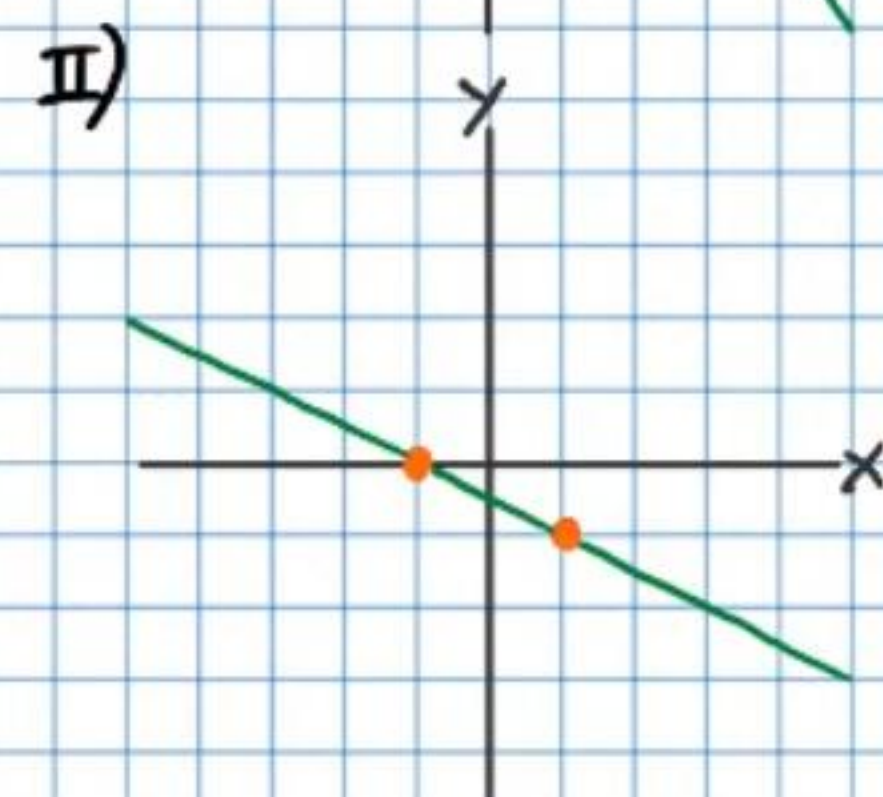
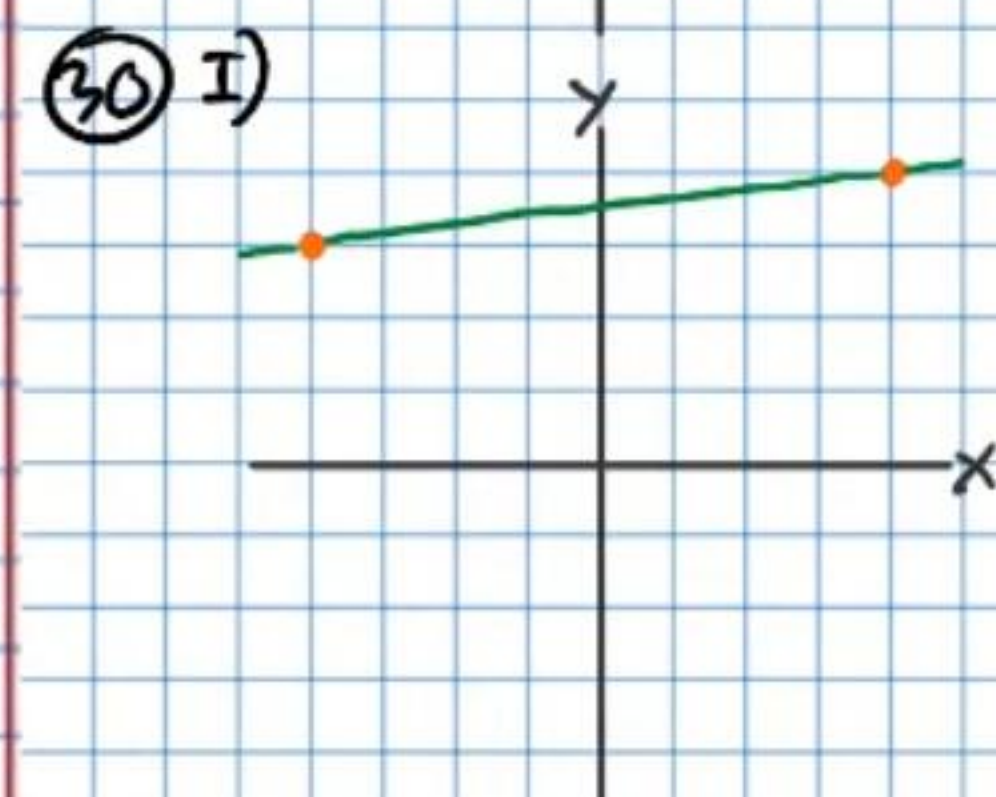
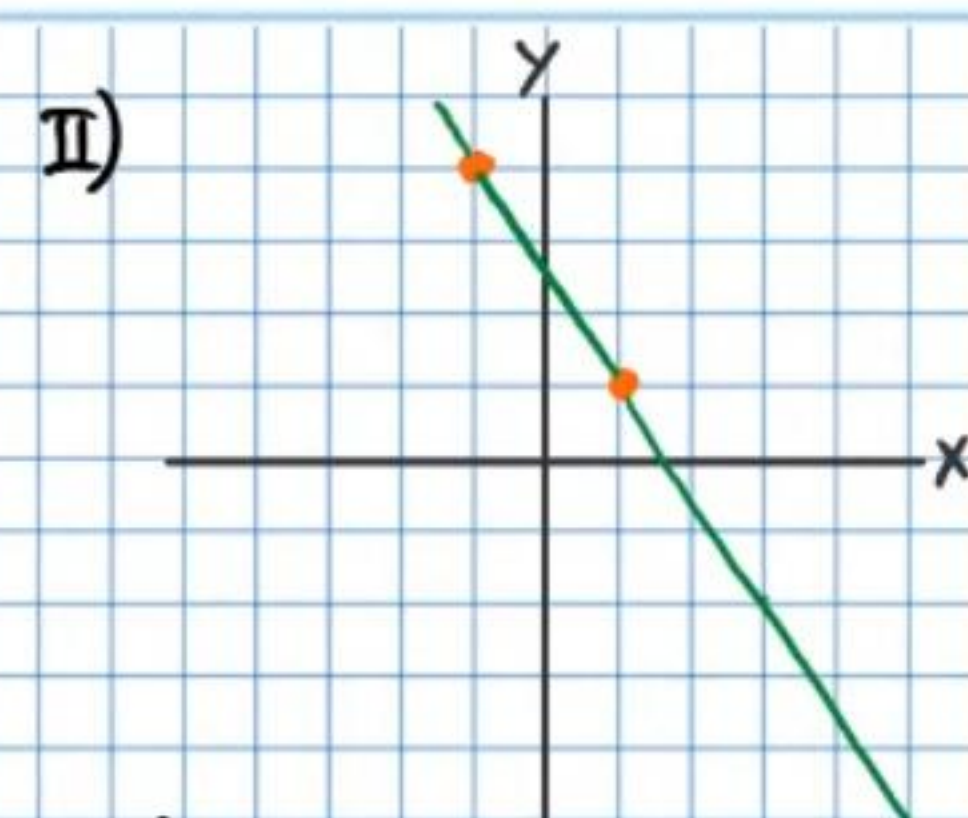
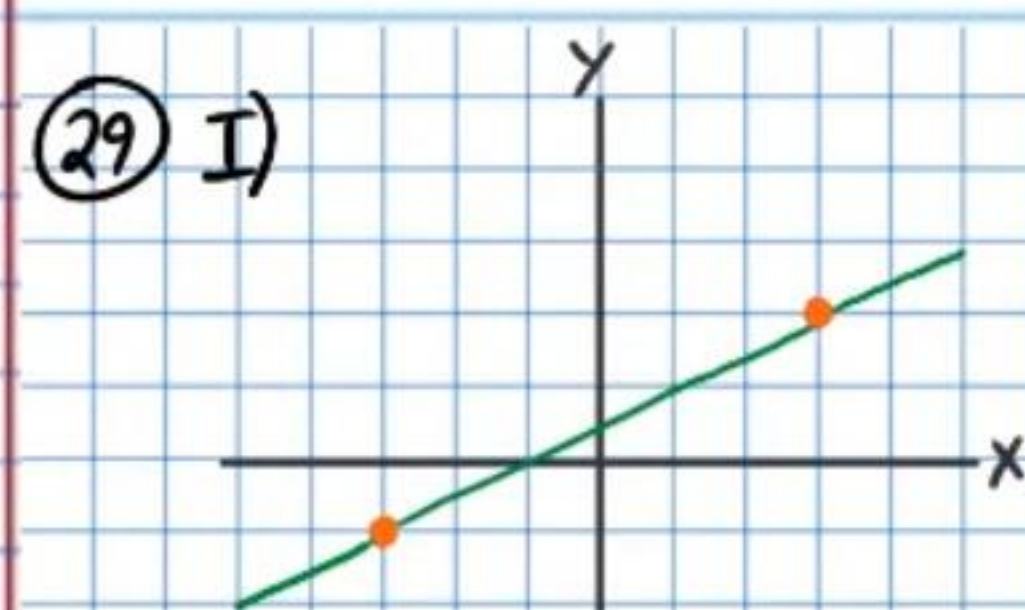
㉔ $\frac{2}{5}(x - 6) = y - 4$

㉕ $y - 2 = -\frac{6}{7}(x + 7)$

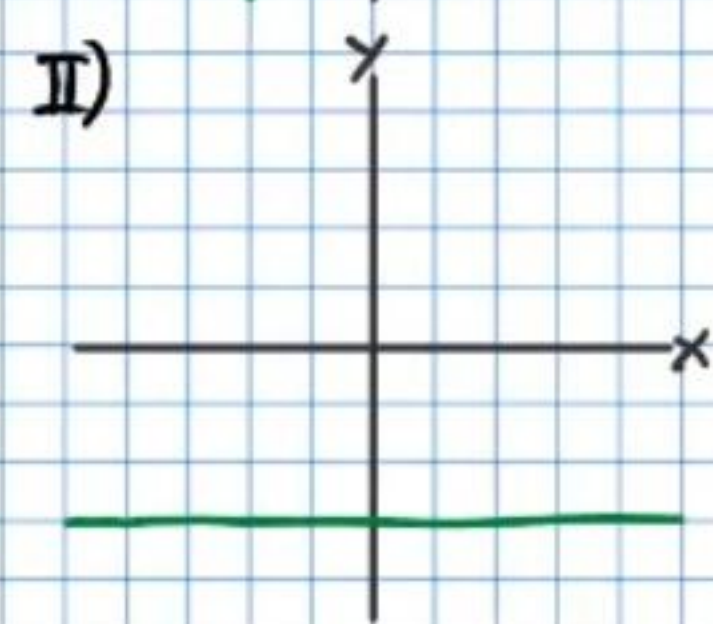
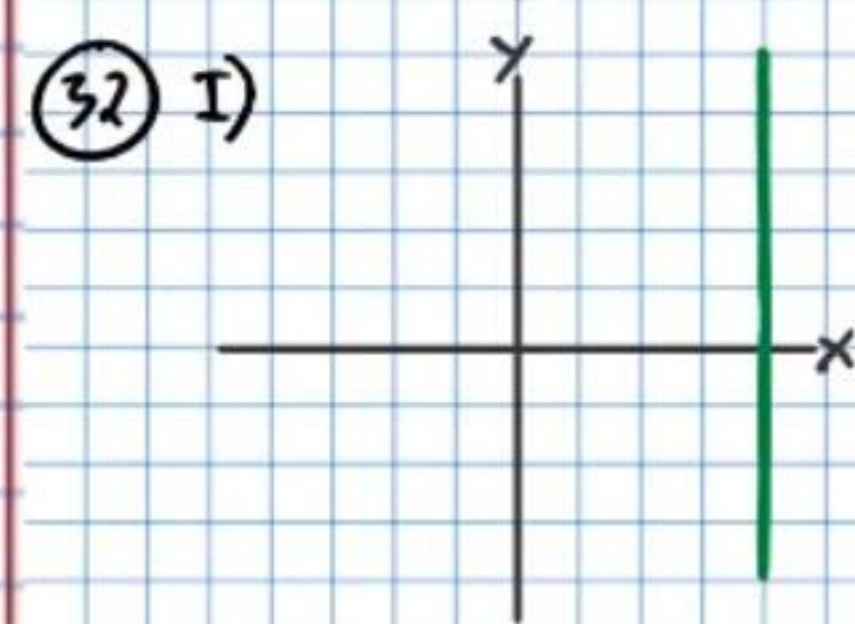
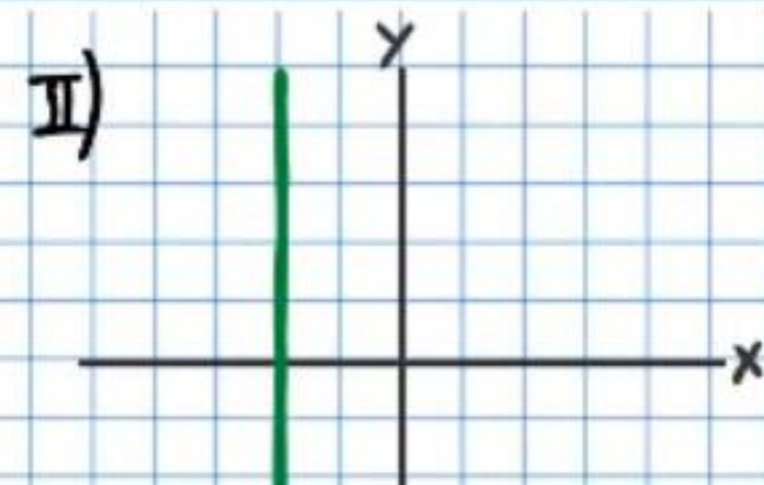
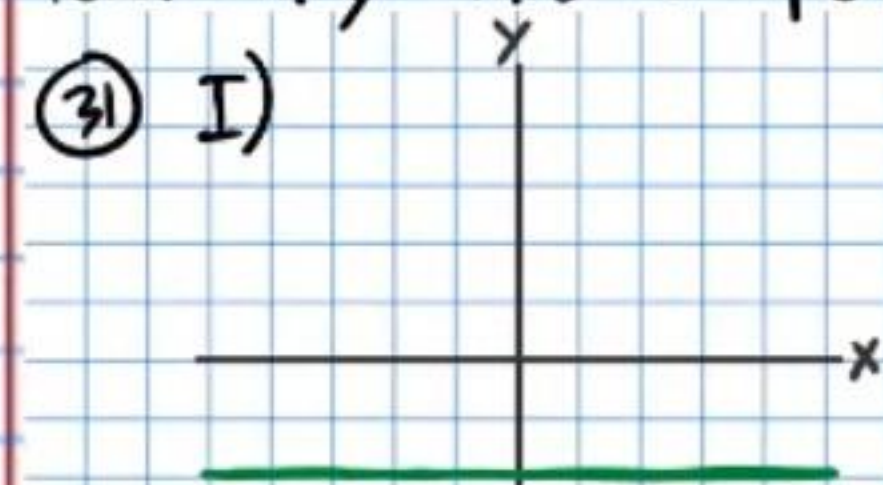
㉖ $-(x + 5) = y + 3$

For each graph below, write an equation of the line shown.

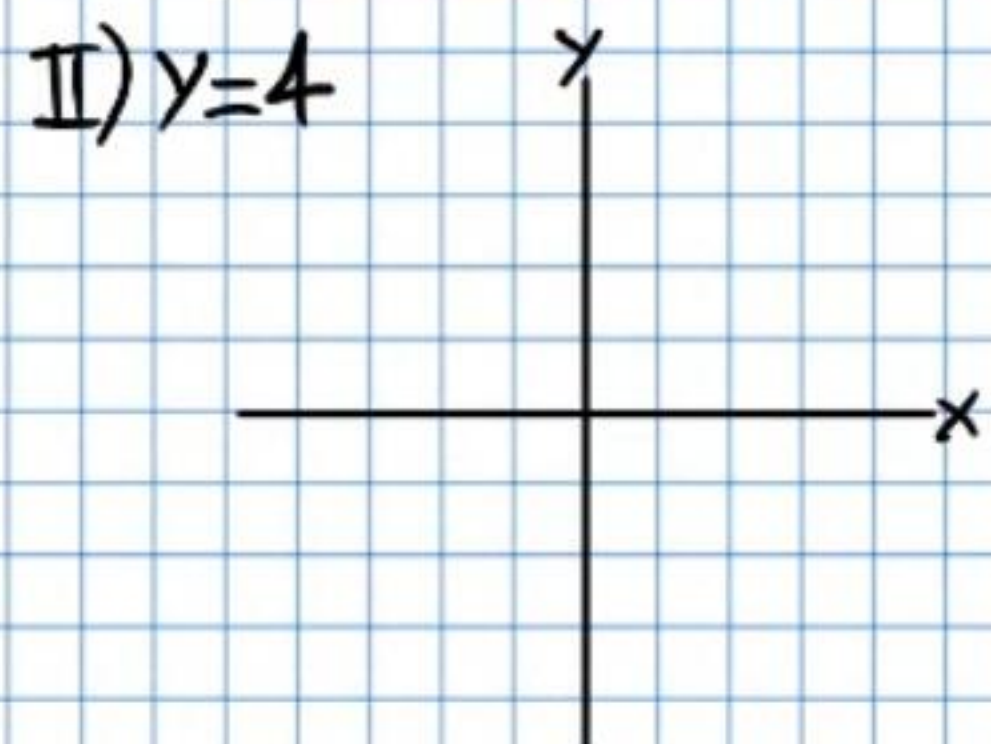
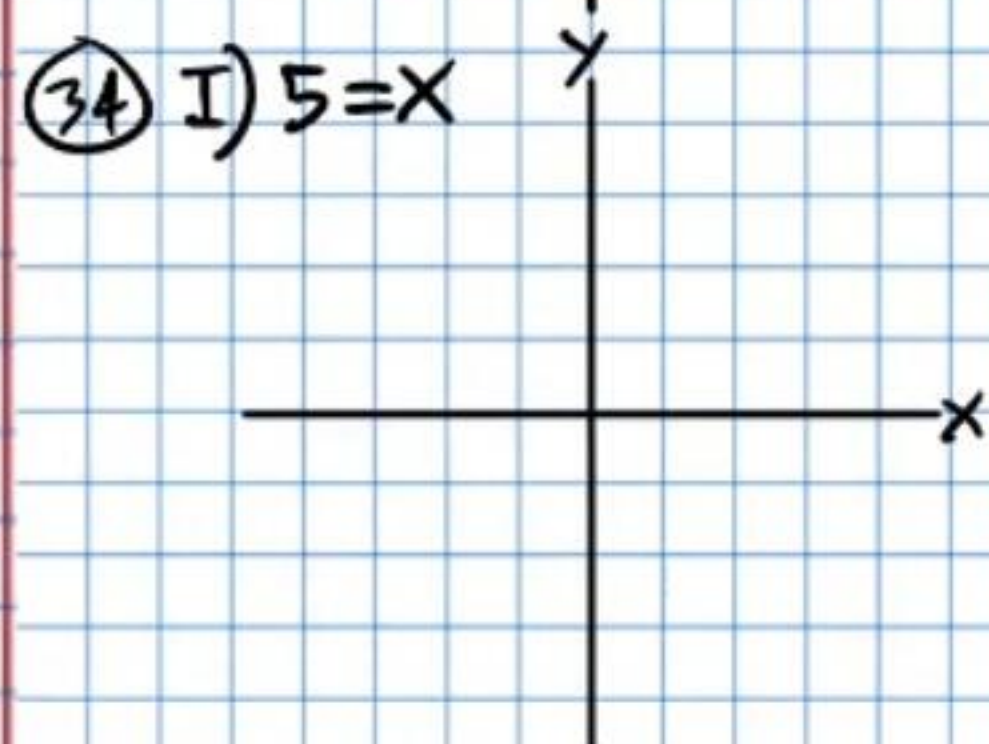
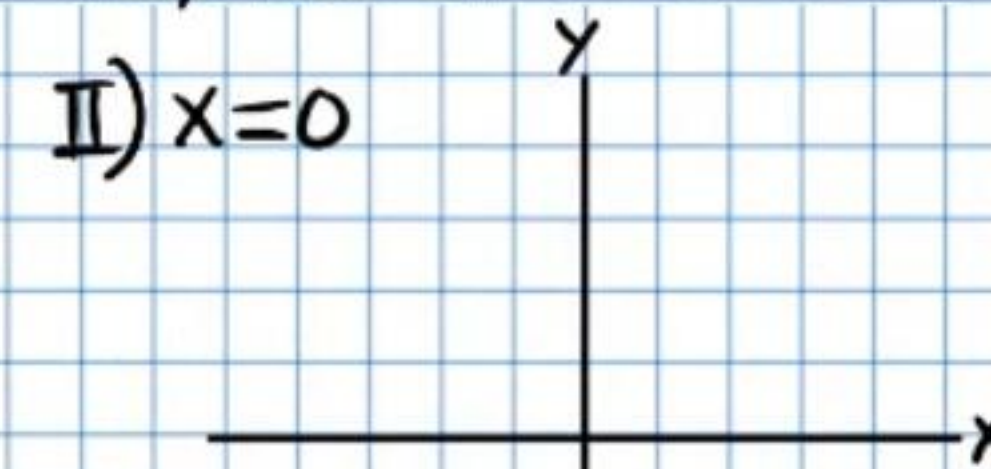
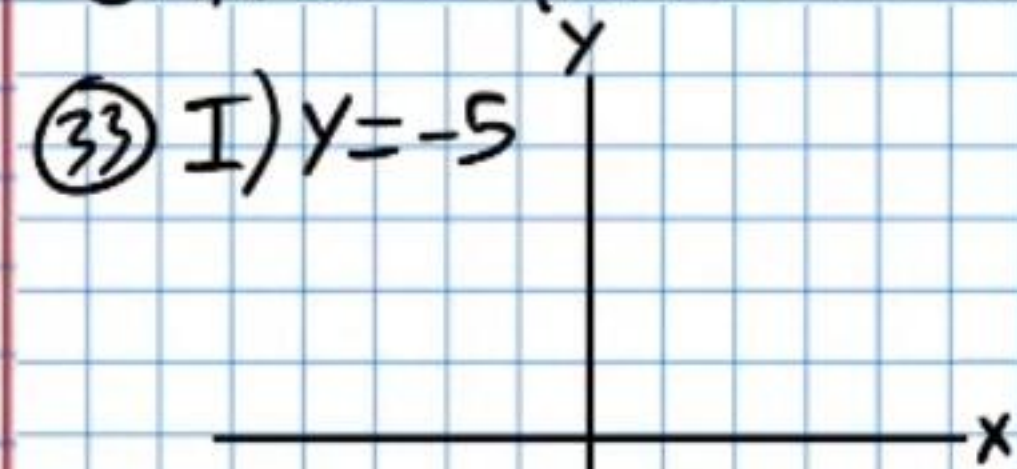




For each graph below, write an equation of the line and identify its slope.



Graph each equation below and identify the slope of the line produced.



Solve the following systems of equations with the substitution method. You must solve for y in one of the equations before doing this.

(35)
$$\begin{cases} y + 3 = 4(x - 2) \\ 5(x - 2) = y - 1 \end{cases}$$

$$\begin{cases} x = -2 \\ y = -19 \end{cases}$$

(36)
$$\begin{cases} -2(x - 3) = y - 2 \\ 3(x + 1) = y + 5 \end{cases}$$

$$\begin{cases} x = 2 \\ y = 4 \end{cases}$$

(37)
$$\begin{cases} y - 8 = 4(x + 2) \\ y + 1 = 2(x - 1) \end{cases}$$

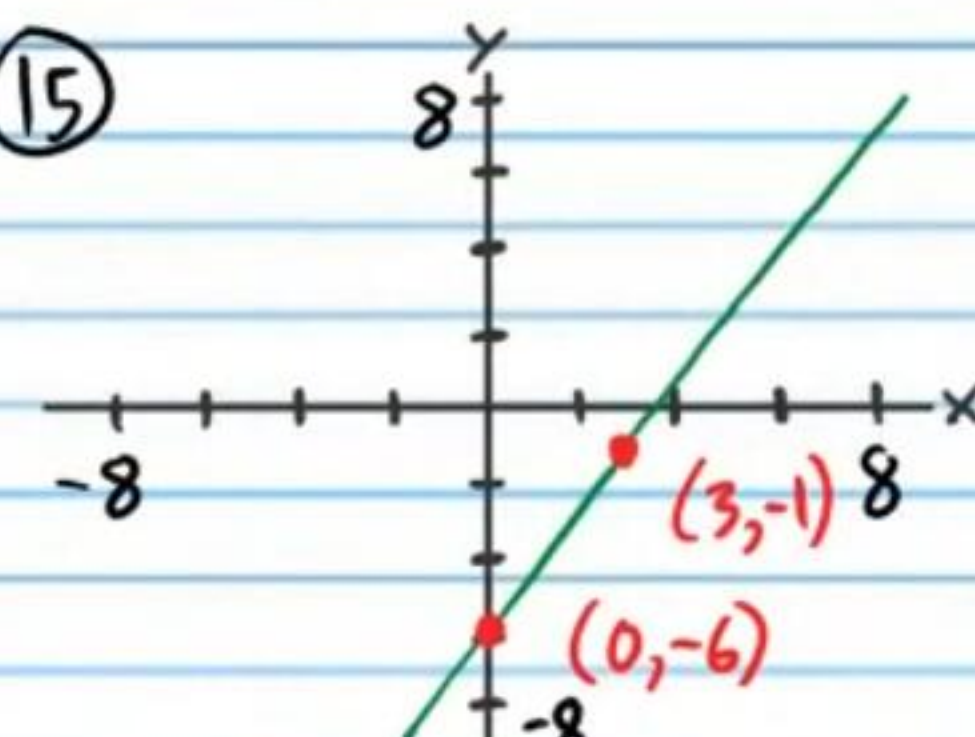
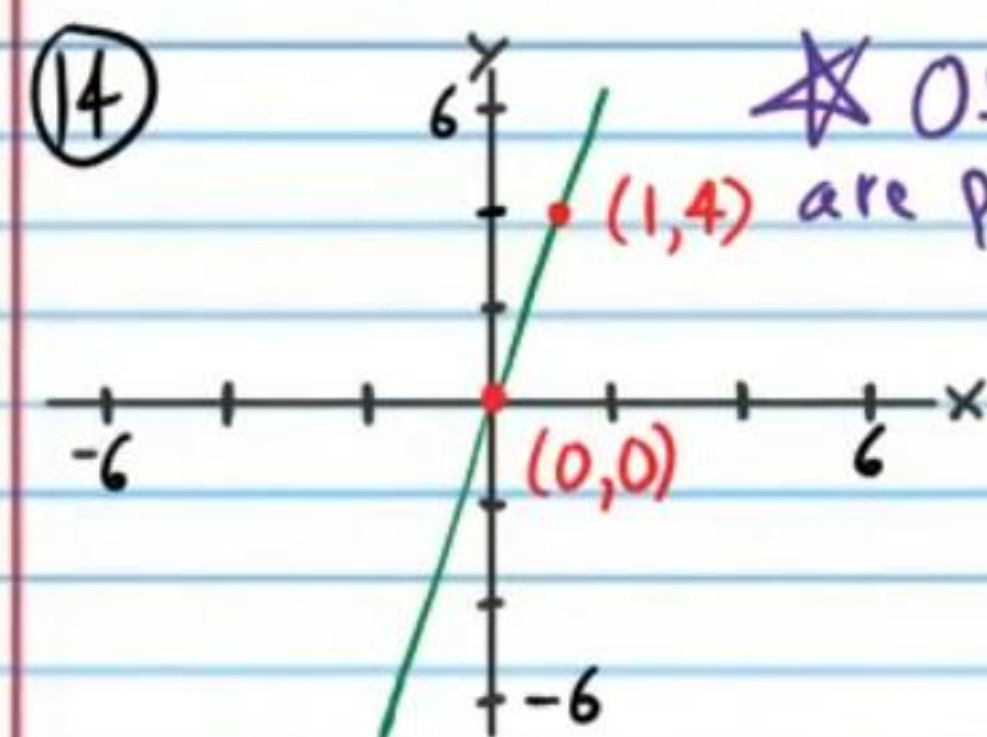
$$\begin{cases} x = -19/2 \\ y = -22 \end{cases}$$

(38)
$$\begin{cases} 3(x + 2) = y + 7 \\ -4(x - 3) = y - 2 \end{cases}$$

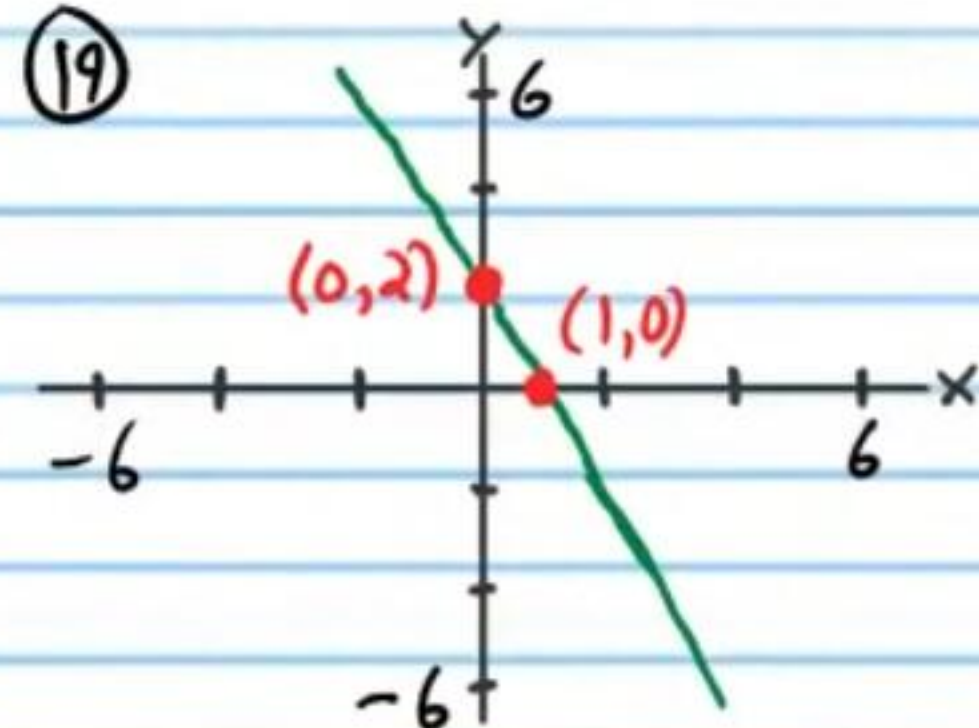
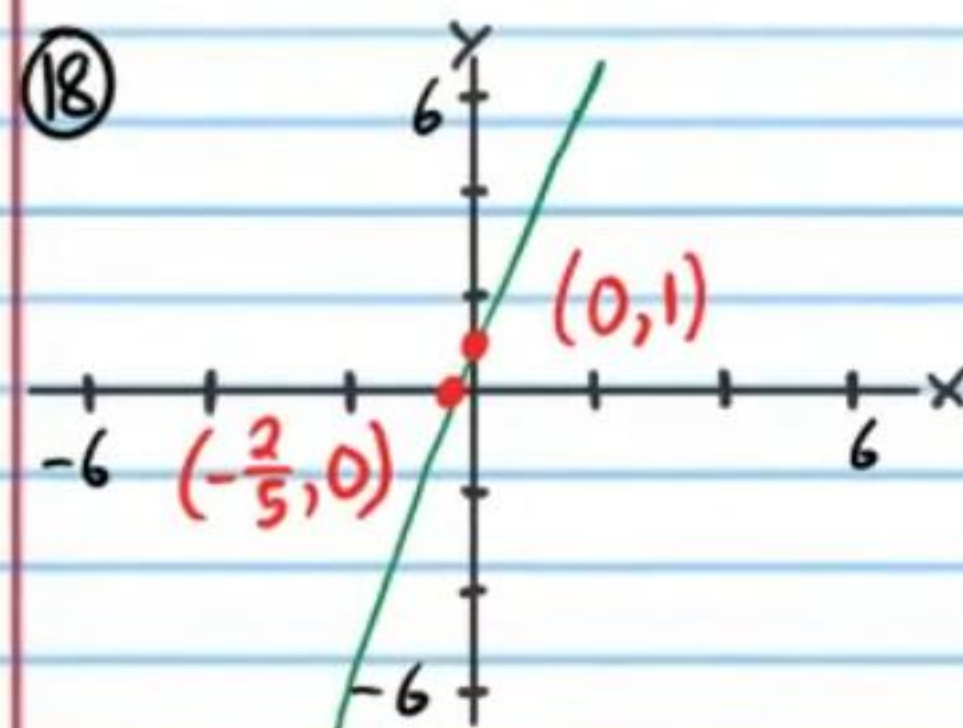
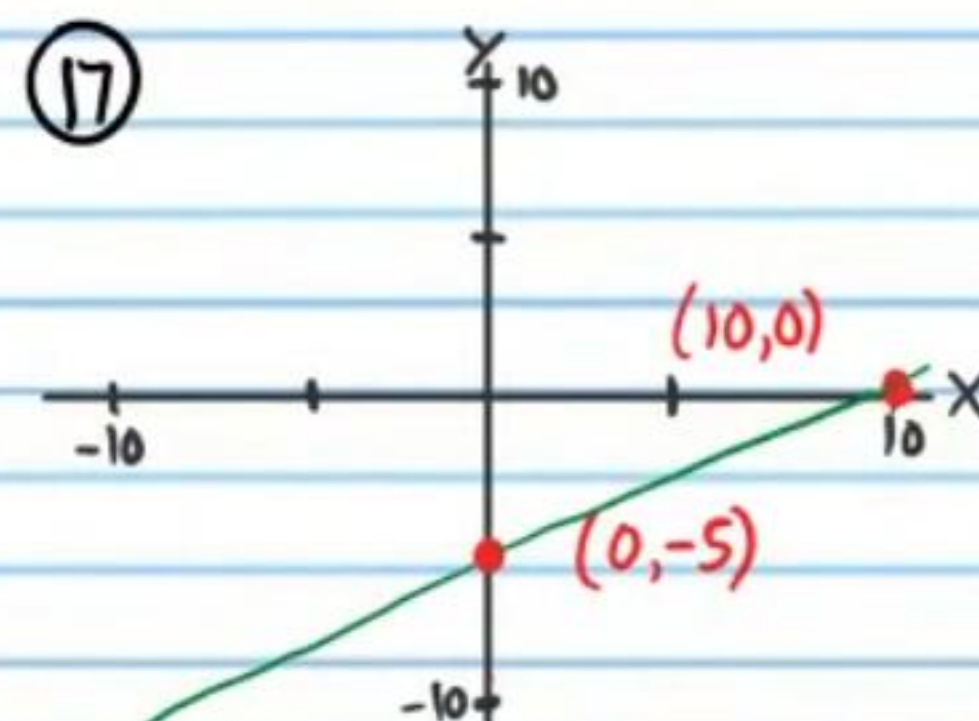
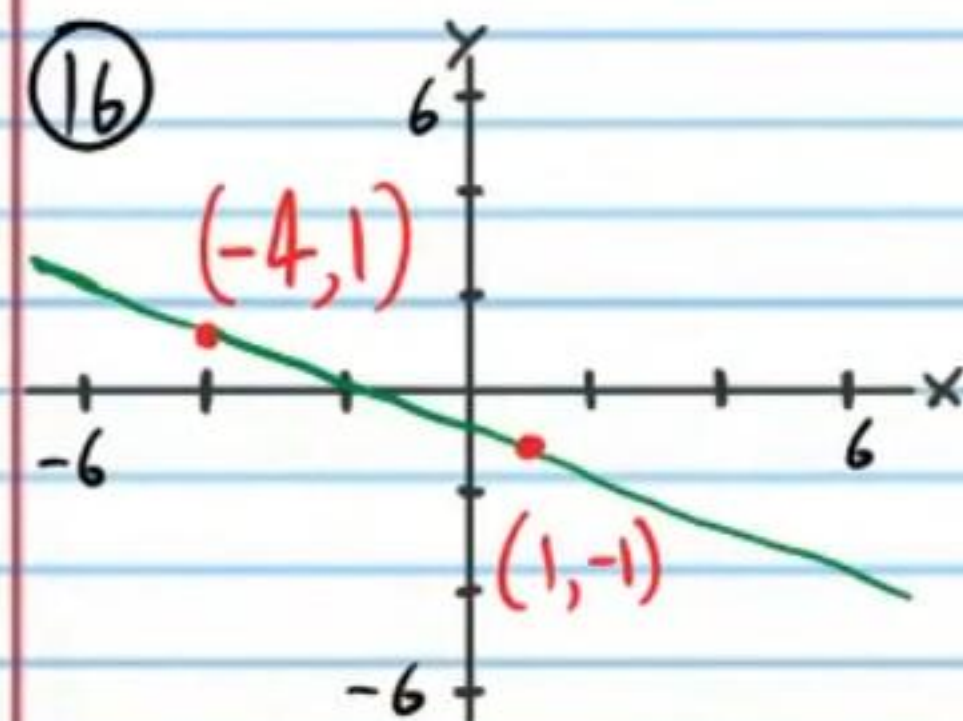
$$\begin{cases} x = 15/7 \\ y = 38/7 \end{cases}$$

Answers for Problem #'s 14-34

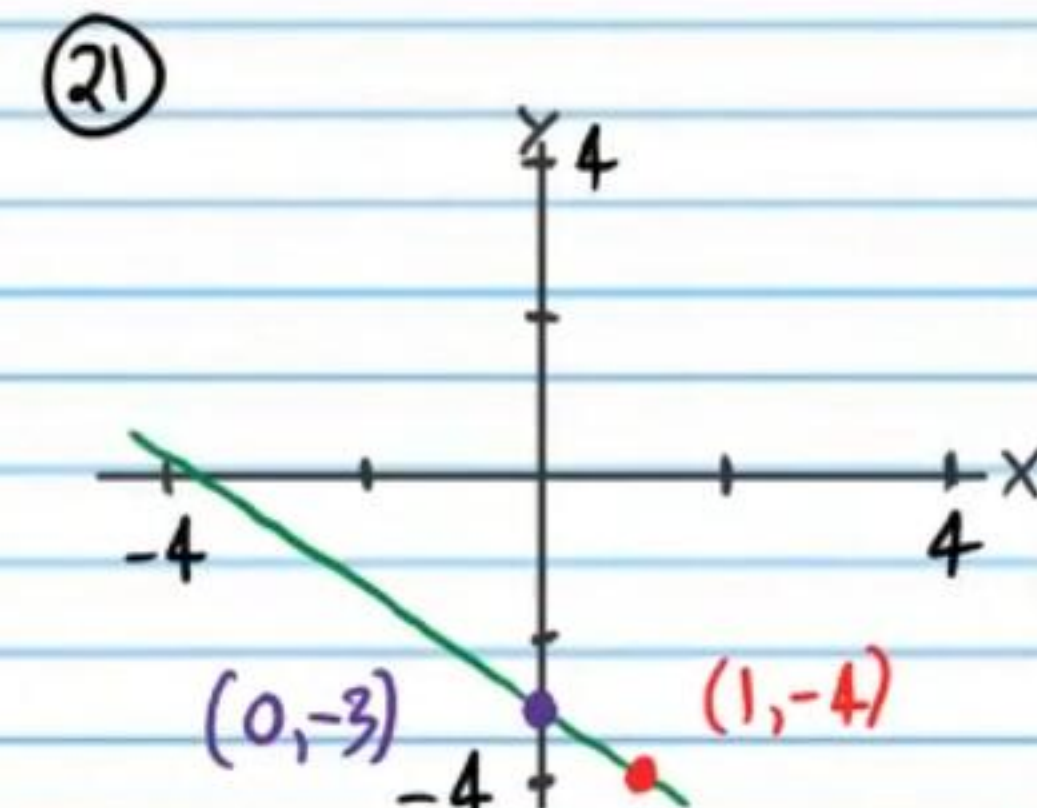
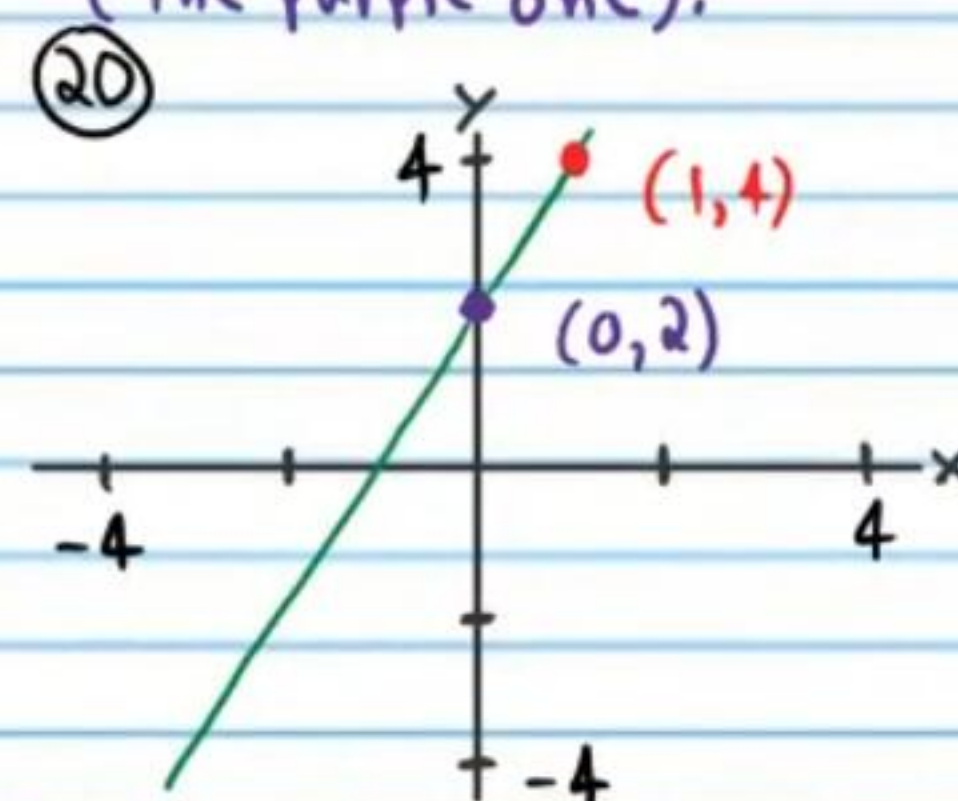
⑭ $\star$ Other points ⑮



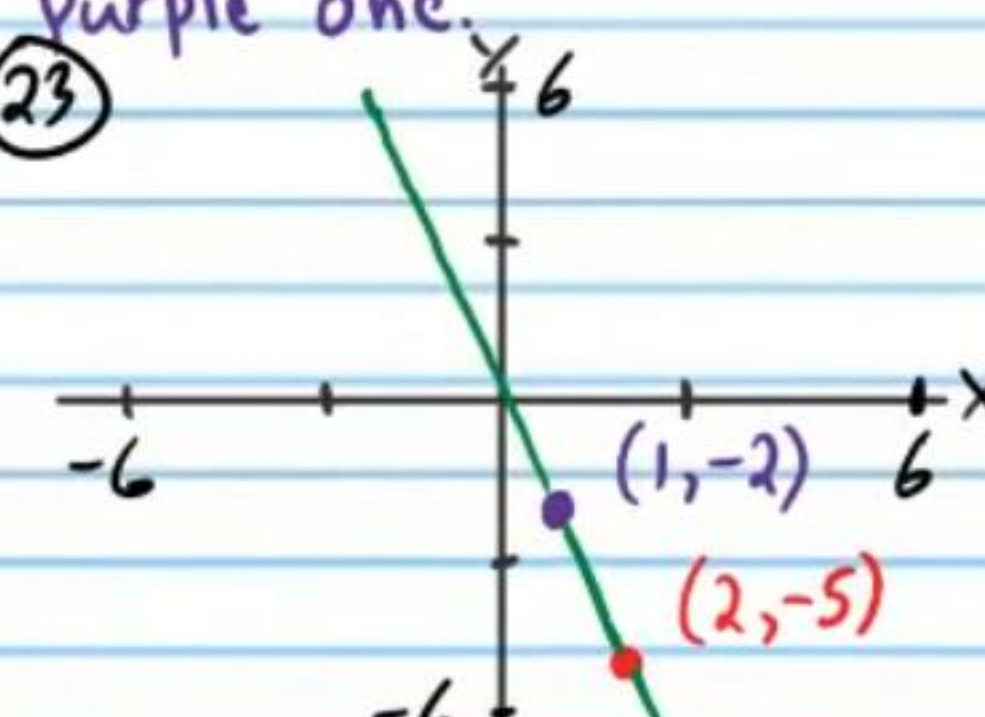
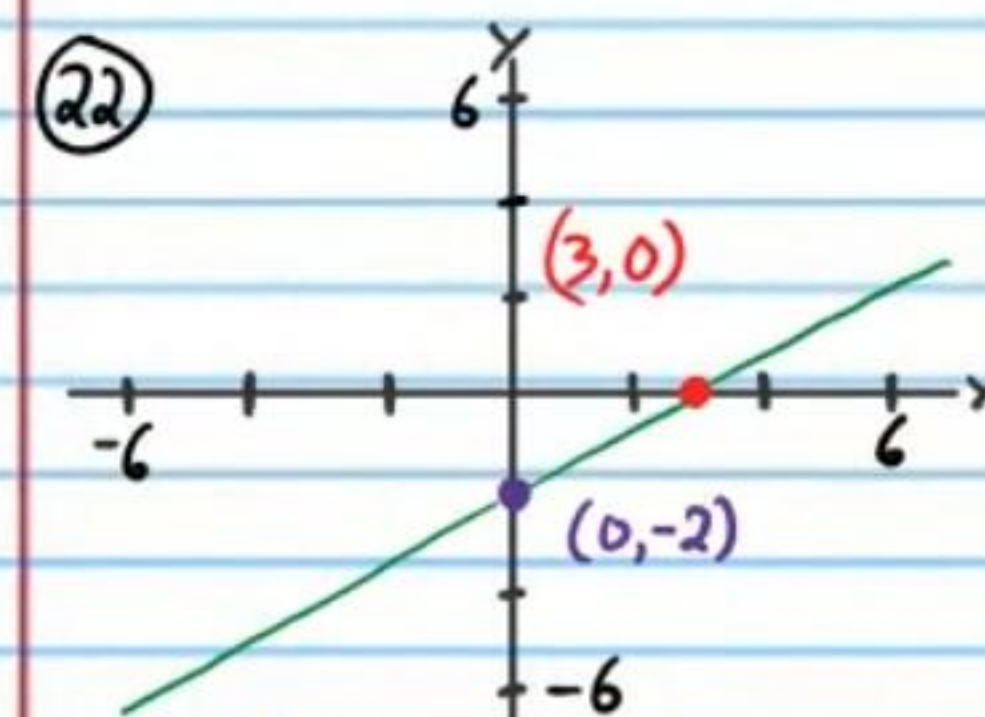
For problem #'s 17-19, one point must be on the x-axis and the other point must be on the y-axis.



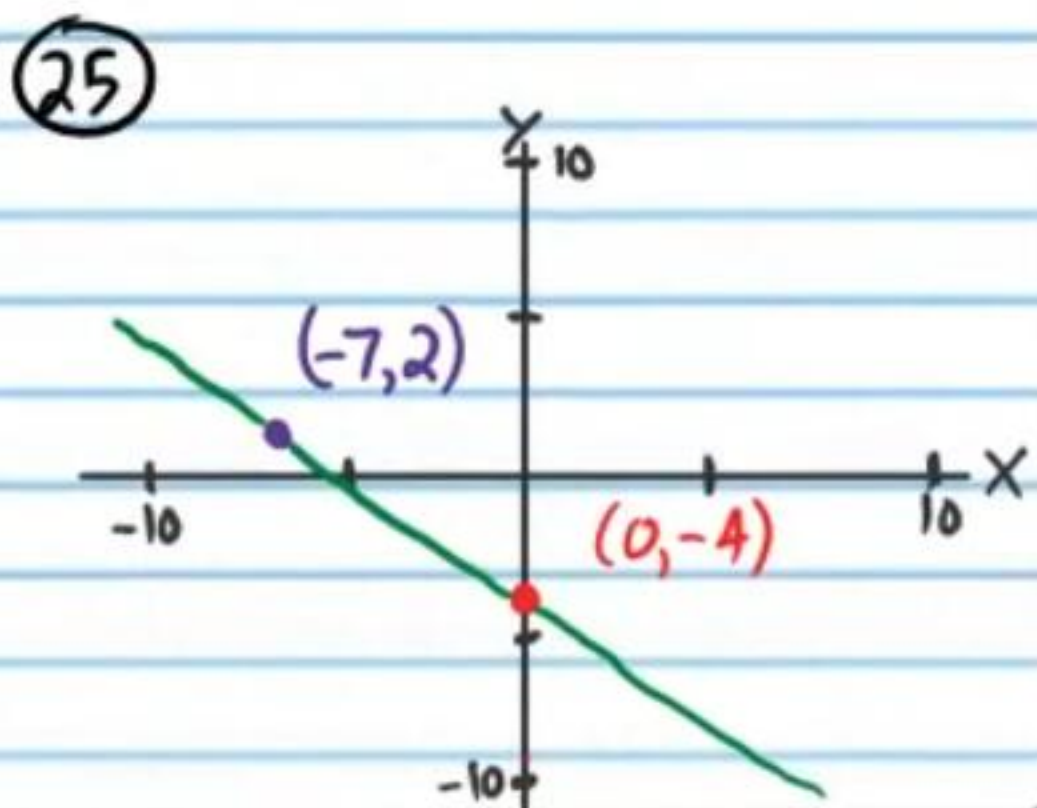
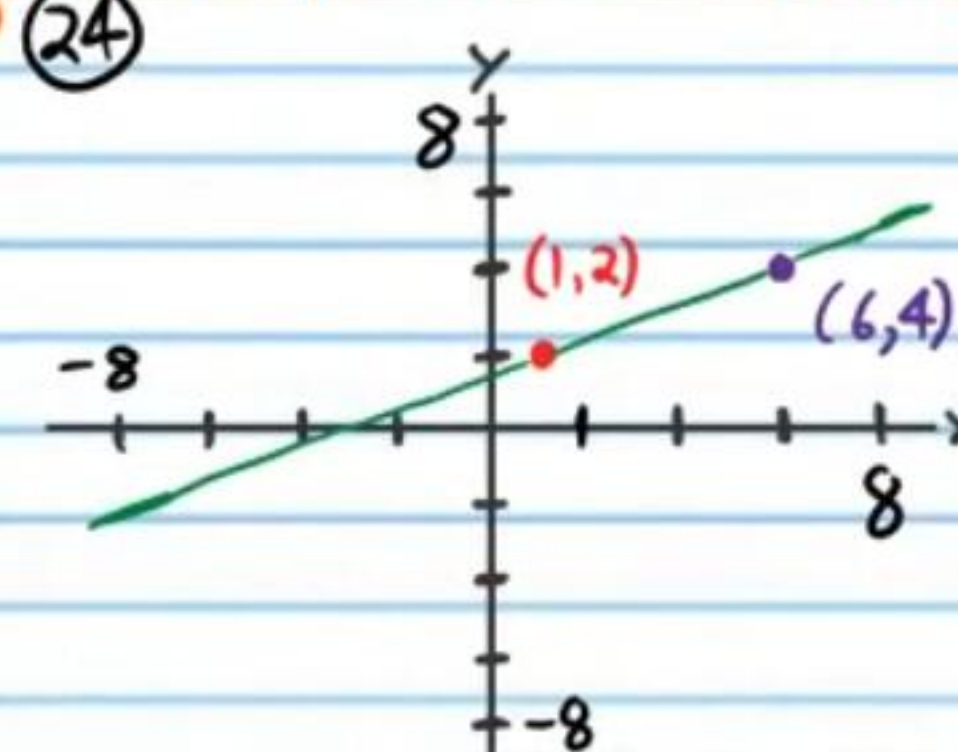
For problem #'s 20-22, one of your points must be on the y-axis (the purple one).

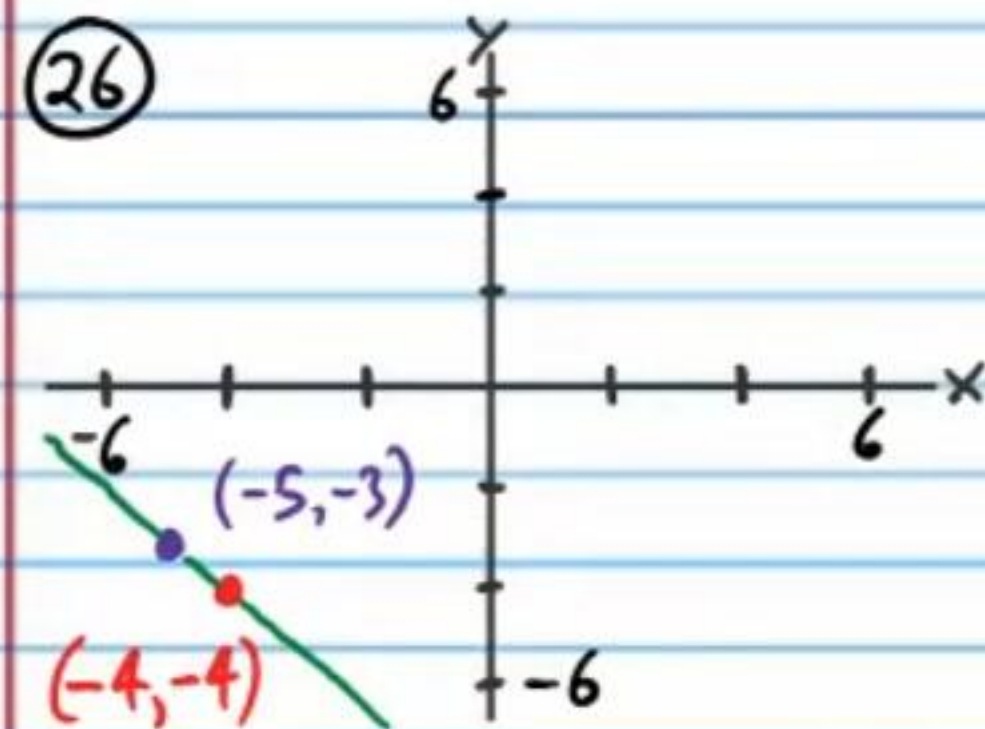


For problem #'s 23-26, one of your points must be the purple one.



$\star$ Moved down and to the left to stay in the window, rather than up and to the right.





★ For problem #'s 27-30, other forms are possible but these are the easiest to find with the given information.

②⑦ I) $y = \frac{5}{4}x + 4$

II) $y = -5x + 1$

②⑧ I) $y = -2x - 2$

II) $y = \frac{4}{3}x - 1$

②⑨ I) $y + 1 = \frac{1}{2}(x + 3)$ or $y - 2 = \frac{1}{2}(x - 3)$

II) $y - 4 = -\frac{3}{2}(x + 1)$ or $y - 1 = -\frac{3}{2}(x - 1)$

③⑩ I) $y - 3 = \frac{1}{8}(x + 4)$ or $y - 4 = \frac{1}{8}(x - 4)$

II) $y - 0 = -\frac{1}{2}(x + 1)$ or $y + 1 = -\frac{1}{2}(x - 1)$

③① I) $y = -2$ $m = 0$

II) $x = -2$ $m = \text{und}$

③② I) $x = 4$ $m = \text{und}$

II) $y = -3$ $m = 0$

